

Create authentication service that returns JWT

1. Create Authentication Controller:

Code:

AuthenticationController.java

```
package com.cognizant.spring_learn.controller;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class AuthenticationController {

    @GetMapping("/authenticate")
    public String authenticate() {
        return "Authenticated Successfully";
    }

}
```

SecurityConfig.java

```
package com.cognizant.spring_learn.config;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.security.config.Customizer;
import org.springframework.security.config.annotation.web.builders.HttpSecurity;
import org.springframework.security.core.userdetails.User;
import org.springframework.security.core.userdetails.UserDetails;
import org.springframework.security.provisioning.InMemoryUserDetailsManager;
import org.springframework.security.web.SecurityFilterChain;

@Configuration
public class SecurityConfig {

    @Bean
    public InMemoryUserDetailsManager userDetailsService() {

        UserDetails user = User
            .withUsername("user")
            .password("{noop}pwd")
            .roles("USER")
            .build();

        return new InMemoryUserDetailsManager(user);
    }

    @Bean
    public SecurityFilterChain securityFilterChain(HttpSecurity http)
```

```
        throws Exception {  
  
        http  
            .authorizeHttpRequests(auth -> auth  
                .requestMatchers("/authenticate").authenticated()  
                .anyRequest().permitAll()  
            .httpBasic(Customizer.withDefaults())  
            .csrf(csrf -> csrf.disable());  
  
        return http.build();  
    }  
}
```

Testing the Authentication:

```
Microsoft Windows [Redacted]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\divit>curl -s -u user:pwd http://localhost:8090/authenticate  
Authenticated Successfully  
C:\Users\divit>
```

2. Read Authorization Header:

Code:

AuthenticationController.java

```
package com.cognizant.spring_learn.controller;

import java.nio.charset.Charset;
import java.util.Base64;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestHeader;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class AuthenticationController {

    @GetMapping("/authenticate")
    public String authenticate(
        @RequestHeader("Authorization") String authHeader) {

        String encodedCredentials =
            authHeader.substring("Basic ".length());

        byte[] decodedBytes =
            Base64.getDecoder().decode(encodedCredentials);

        String credentials =
            new String(decodedBytes, Charset.forName("UTF-8"));

        String username =
            credentials.split(":")[0];

        String password =
            credentials.split(":")[1];

        return "Username : " + username +
            " Password : " + password;
    }
}
```

Testing the Authentication:

```
Microsoft Windows [Version 6.0.6002.18005]
(c) Microsoft Corporation. All rights reserved.

C:\Users\divit>curl -s -u user:pwd http://localhost:8090/authenticate
Username : user Password : pwd
C:\Users\divit>
```

3. Generate JWT Token

Code:

SecurityConfig.java – Config Package

```
package com.cognizant.spring_learn.config;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.security.config.Customizer;
import org.springframework.security.config.annotation.web.builders.HttpSecurity;
import org.springframework.security.core.userdetails.User;
import org.springframework.security.core.userdetails.UserDetails;
import org.springframework.security.provisioning.InMemoryUserDetailsManager;
import org.springframework.security.web.SecurityFilterChain;

@Configuration
public class SecurityConfig {

    @Bean
    public InMemoryUserDetailsManager userDetailsService() {

        UserDetails user = User
            .withUsername("user")
            .password("{noop}pwd")
            .roles("USER")
            .build();

        return new InMemoryUserDetailsManager(user);
    }
}
```

```

@Bean
public SecurityFilterChain securityFilterChain(HttpSecurity http)
    throws Exception {

    http
        .authorizeHttpRequests(auth -> auth
            .requestMatchers("/authenticate").authenticated()
            .anyRequest().permitAll())
        .httpBasic(Customizer.withDefaults())
        .csrf(csrf -> csrf.disable());

    return http.build();
}
}

```

AuthenticationController.java – Controller Package

```

package com.cognizant.spring_learn.controller;

import java.nio.charset.Charset;
import java.util.Base64;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestHeader;
import org.springframework.web.bind.annotation.RestController;

import com.cognizant.spring_learn.model.AuthenticationResponse;
import com.cognizant.spring_learn.util.JwtUtil;

@RestController
public class AuthenticationController {

    @GetMapping("/authenticate")
    public AuthenticationResponse authenticate(
        @RequestHeader("Authorization")
        String authHeader) {

        String encodedCredentials =
            authHeader.substring("Basic ".length());

        byte[] decodedBytes =
            Base64.getDecoder()
                .decode(encodedCredentials);

        String credentials =
            new String(decodedBytes,
                Charset.forName("UTF-8"));

        String username =
            credentials.split(":")[0];

        String token =
            JwtUtil.generateToken(username);
    }
}

```

```

        return new AuthenticationResponse(token);
    }
}

```

AuthenticationResponse.java – Model Package

```

package com.cognizant.spring_learn.model;

public class AuthenticationResponse {

    private String token;

    public AuthenticationResponse() {
    }

    public AuthenticationResponse(String token) {
        this.token = token;
    }

    public String getToken() {
        return token;
    }

    public void setToken(String token) {
        this.token = token;
    }
}

```

JwtUtil.java – Util Package

```

package com.cognizant.spring_learn.util;

import java.util.Date;

import io.jsonwebtoken.Jwts;
import io.jsonwebtoken.SignatureAlgorithm;

public class JwtUtil {

    private static final String SECRET_KEY = "secretkey";

    public static String generateToken(String username) {

        return Jwts.builder()
            .setSubject(username)
            .setIssuedAt(new Date())
            .setExpiration(
                new Date(System.currentTimeMillis() + 1200000))
            .signWith(SignatureAlgorithm.HS256, SECRET_KEY)
            .compact();
    }
}

```

```
}  
}
```

Output:

```
Microsoft Windows [redacted]  
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C:\Users\divit>curl -s -u user:pwd http://localhost:8090/authenticate  
{  
  "token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJ1c2VyIiwiaWF0IjoxNzg0NjQyMzcwLCJleHAiOiE3ODQ2NDM1NzF9.ArPbM8i51byDnL1FhSOcbtHWXy10JFHYKn-6iohkjx0"  
}  
C:\Users\divit>
```